

## Sequence Detector

ex: if we have sequence (let 10110110001011--)  
and if we see specific sequence (let 1011) make action

### Sequence Detector

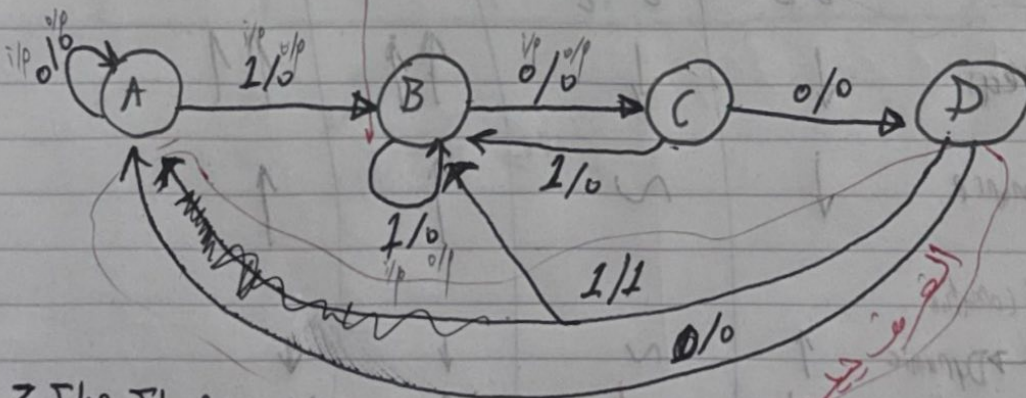
→ over Lapping

ex: 1011011

→ Non over lapping

ex: 1011 011 000 1011 1

Ex: -- 1100111 over lapping Sequence Detector Mealy Machine



2 Flip Flop  
2 i/p  
1 o/p



$A \rightarrow 00$   
 $B \rightarrow 01$   
 $C \rightarrow 10$   
 $D \rightarrow 11$

| $Q_A Q_B$ | 00 | 01 | 11 | 10 |
|-----------|----|----|----|----|
| X         | 0  | 1  | 1  | 1  |
|           | 1  | 1  | 1  | 1  |

$Q_B^+$

$$Q_B^+ = D_B = X + Q_A Q_B'$$

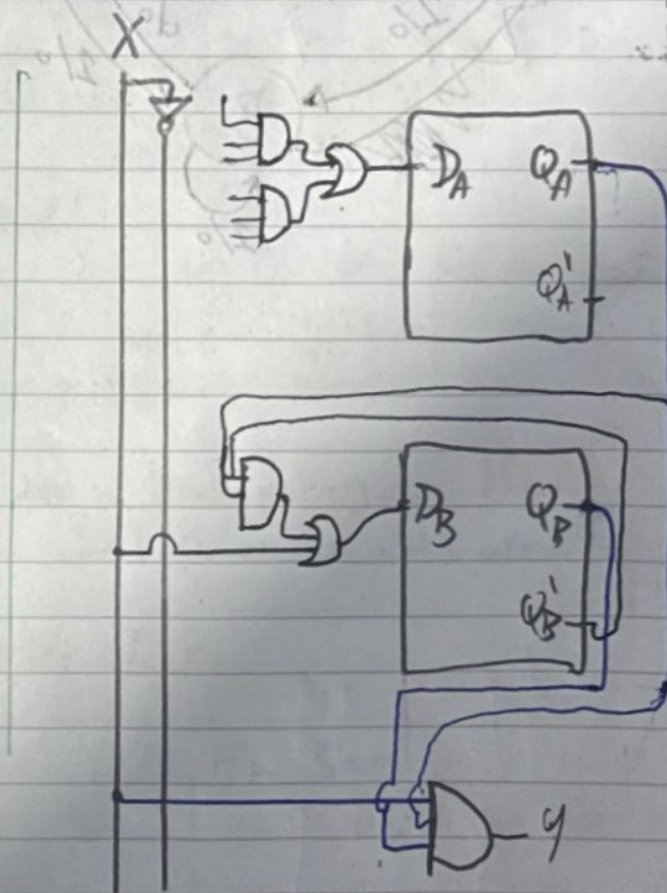
| $Q_A Q_B$ | 00 | 01 | 11 | 10 |
|-----------|----|----|----|----|
| X         | 0  | 1  |    | 1  |
|           | 1  |    |    |    |

$Q_A^+$

$$D_A = Q_A^+ = Q_A' Q_B X' + Q_A Q_B' X'$$

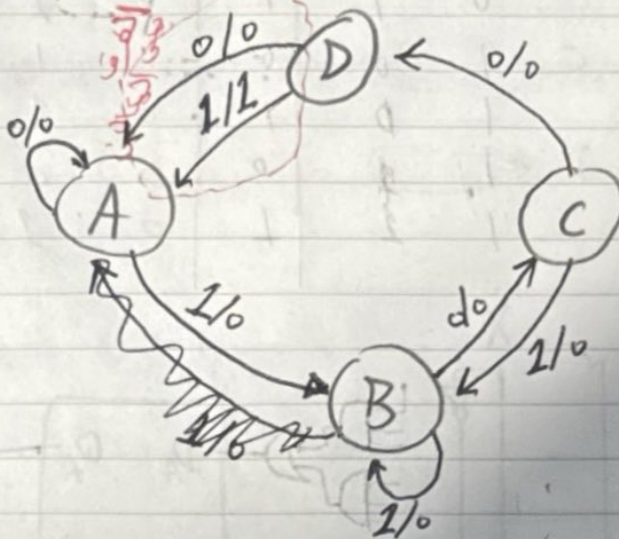
$$y = Q_A Q_B X$$

|   | $Q_A$ | $Q_B$ | X | $Q_A^+$ | $Q_B^+$ | y |
|---|-------|-------|---|---------|---------|---|
| A | 0     | 0     | 0 | 0       | 0       | 0 |
| " | 0     | 0     | 1 | 0       | 1       | 0 |
| B | 0     | 1     | 0 | 1       | 0       | 0 |
| " | 0     | 1     | 1 | 0       | 1       | 0 |
| C | 1     | 0     | 0 | 1       | 1       | 0 |
| " | 1     | 0     | 1 | 0       | 1       | 0 |
| D | 1     | 1     | 0 | 0       | 0       | 0 |
| " | 1     | 1     | 1 | 0       | 1       | 1 |



Ex. (1 1001) Non-overlapping sequence Detector Mealy machine

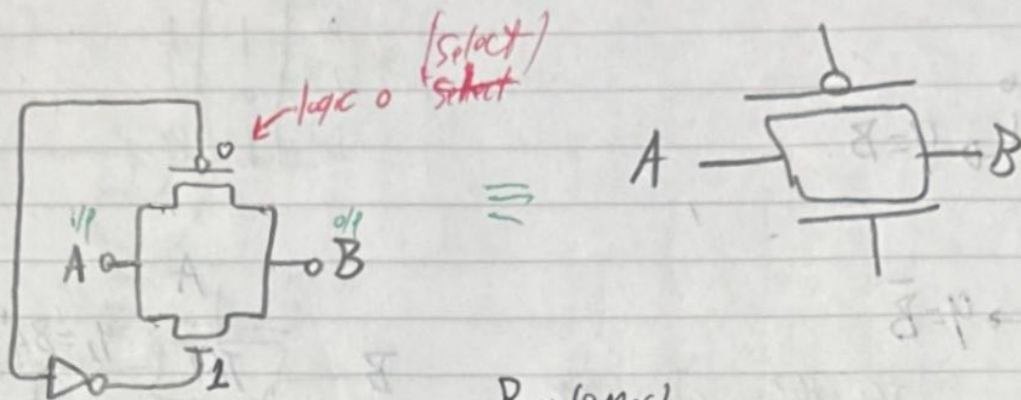
No. of state = num. of state of sequence = 4



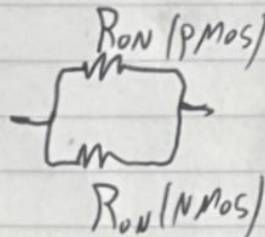


# Transmission Gates

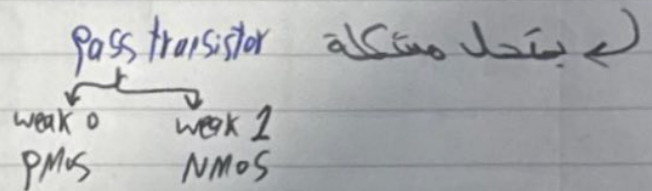
Circuit



Req is small

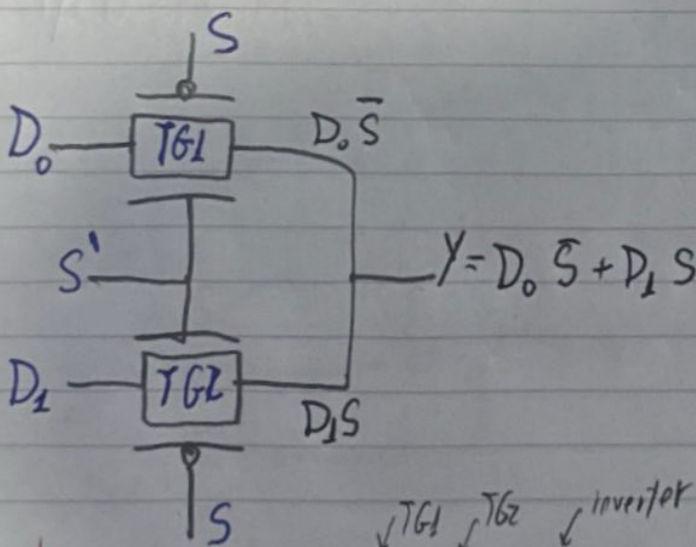


Gives strong '0' & strong '1'



Design Implement  $\rightarrow$  transmission gates  $\rightarrow$  power  $\rightarrow$  area

$\rightarrow$  Implementation 2:1 Mux using T.G



if  $S=0$   
 TG1 is on / TG2 is off  
 $y = D_0 \bar{S}$

if  $S=1$   
 TG1 is off / TG2 is on  
 $y = D_1 S$

$y = D_0 \bar{S} + D_1 S$

$\rightarrow$  No of transistor = 2 + 2 + 2 = 6

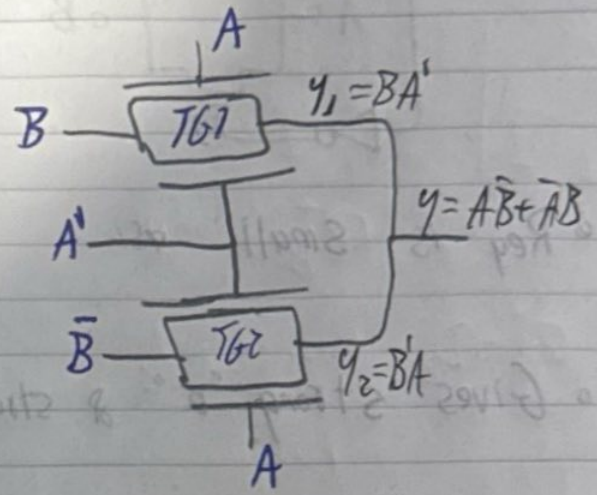
# Design XOR using TG?

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

IF  $A=0$   
 $\hookrightarrow Y=B$

IF  $A=1$   
 $\hookrightarrow Y=\bar{B}$

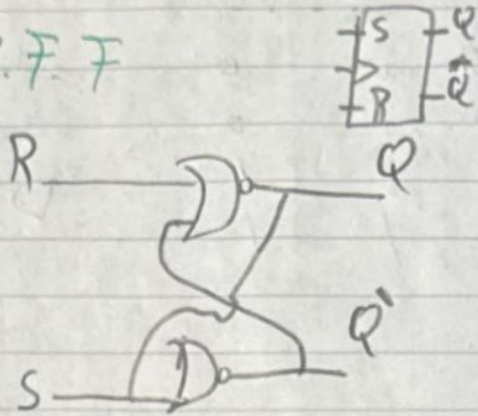
No. of TG = 2 + 2 + 4





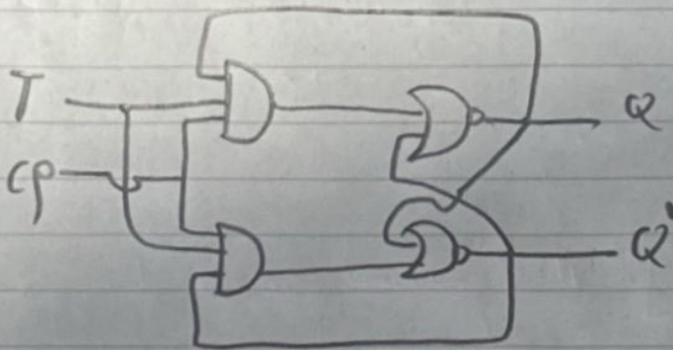
## Flip Flop gate level

→ SR FF



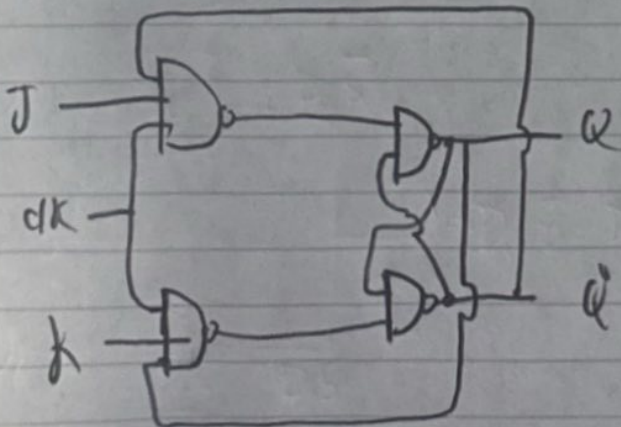
| S | R | Q | Q <sub>next</sub> |           |
|---|---|---|-------------------|-----------|
| 0 | 0 | 0 | 0                 | بنا ما هم |
| 0 | 0 | 1 | 1                 |           |
| 0 | 1 | 0 | 0                 | هناك 0    |
| 0 | 1 | 1 | 0                 | قيمة S    |
| 1 | 0 | 0 | 1                 |           |
| 1 | 0 | 1 | 1                 |           |
| 1 | 1 | 0 |                   | مبطل      |
| 1 | 1 | 1 |                   | يتحقق     |

→ T FF



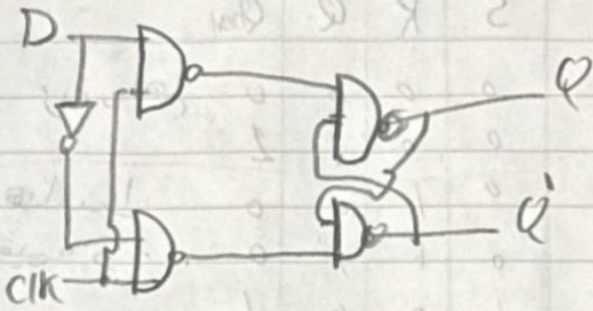
| T | Q | Q <sub>next</sub> |       |
|---|---|-------------------|-------|
| 0 | 0 | 0                 | مبطل  |
| 0 | 1 | 1                 | ما هم |
| 1 | 0 | 1                 | عكس   |
| 1 | 1 | 0                 | عكس   |

→ JK FF



| S | R | Q | Q <sub>next</sub> |            |
|---|---|---|-------------------|------------|
| 0 | 0 | 0 | 0                 | بنا ما هم  |
| 0 | 0 | 1 | 1                 |            |
| 0 | 1 | 0 | 0                 | هناك 0     |
| 0 | 1 | 1 | 0                 | قيمة S     |
| 1 | 0 | 0 | 1                 |            |
| 1 | 0 | 1 | 1                 |            |
| 1 | 1 | 0 | 1                 | عكس قيمة R |
| 1 | 1 | 1 | 0                 |            |

→ D Flip Flop



| D | Q | Q <sub>next</sub> |
|---|---|-------------------|
| 0 | 0 | 0                 |
| 0 | 1 | 0                 |
| 1 | 0 | 1                 |
| 1 | 1 | 1                 |

حالت  
قبله  
حالت  
بعدی  
D

| clk | D | Q |
|-----|---|---|
| 0   | 0 | 0 |
| 1   | 1 | 0 |
| 1   | 0 | 1 |
| 0   | 1 | 1 |

| clk | D | Q | Q <sub>next</sub> |
|-----|---|---|-------------------|
| 0   | 0 | 0 | 0                 |
| 1   | 1 | 0 | 0                 |
| 0   | 0 | 1 | 0                 |
| 0   | 1 | 1 | 0                 |
| 1   | 0 | 0 | 1                 |
| 1   | 1 | 0 | 1                 |
| 0   | 0 | 1 | 1                 |
| 0   | 1 | 1 | 1                 |